

PROJECT: BEE-PC

sub-system: EXP

Version: 1.0.0



Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 01_COVER PAGE.SchDoc	Sheet 1 of 29
Checked by: Bao Bui	Document No: 1	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

1	2	3	4	5	6
A					
B					
C					
D					
1	2	3	4	5	6



Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 02_BLOCK DIAGRAM.Sch	Sheet 2 of 29
Checked by: Bao Bui	Document No: 2	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

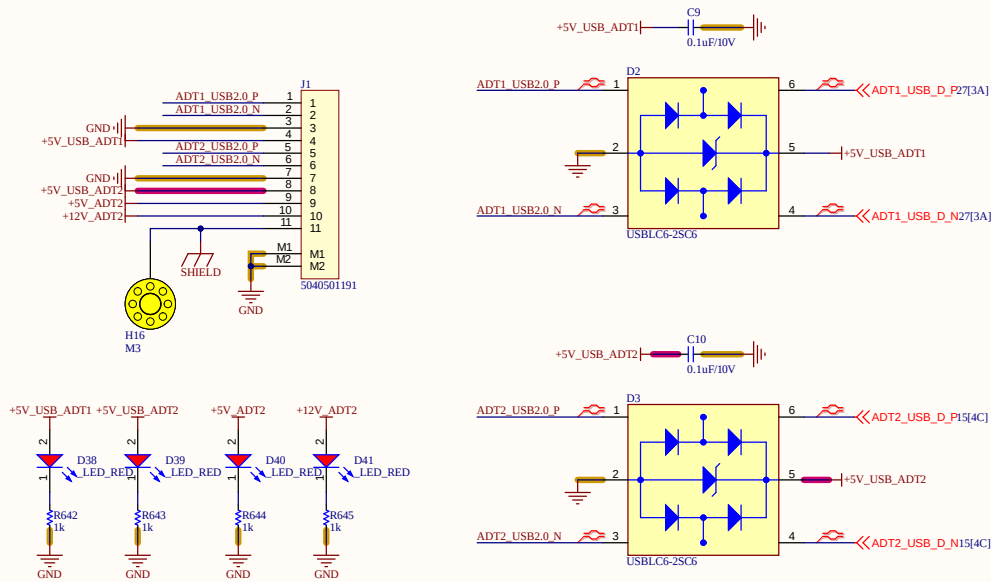
INDEX

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2	BLOCK DIAGRAM	12		22	
3	INDEX	13		23	
4	PI HAT	14		24	
5		15		25	
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7		17		27	
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9		19		29	
10		20		30	



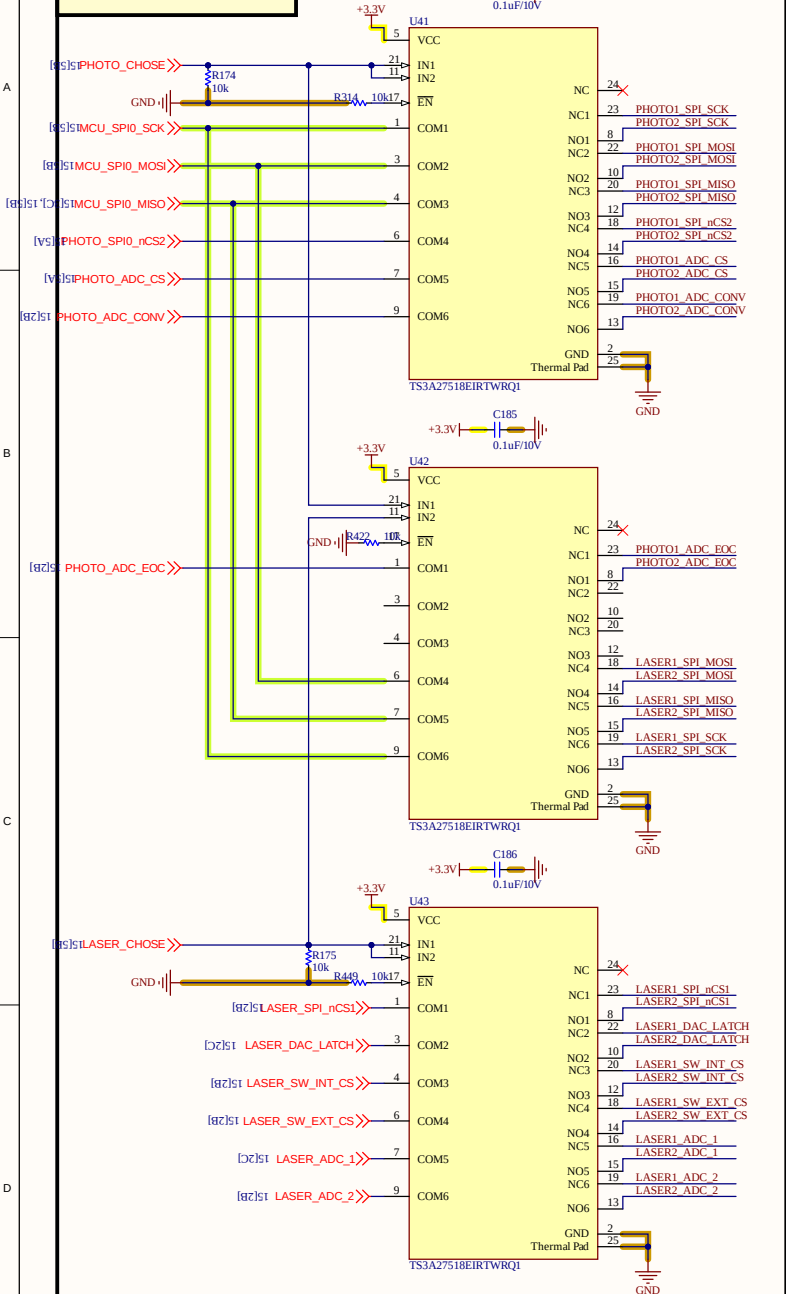
Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 03_INDEX.SchDoc	Sheet 3 of 29
Checked by: Bao Bui	Document No: 3	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

form Adaptor Board

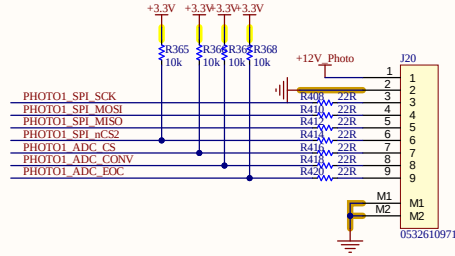


Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 04_CONNECTOR_IN.Sch	Sheet 4 of 29
Checked by: Bao Bui	Document No: 4	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

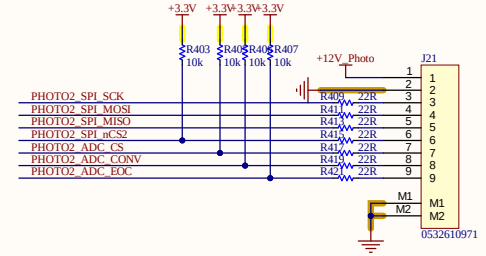
MCU SPI PHOTO



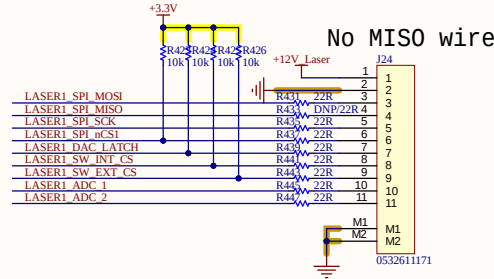
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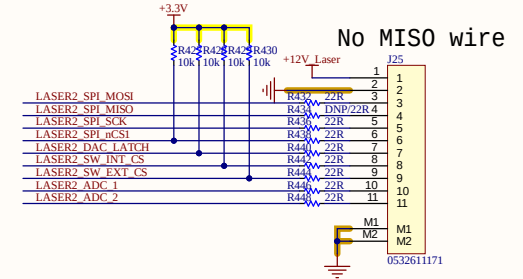
for PHOTO board



for LASER board



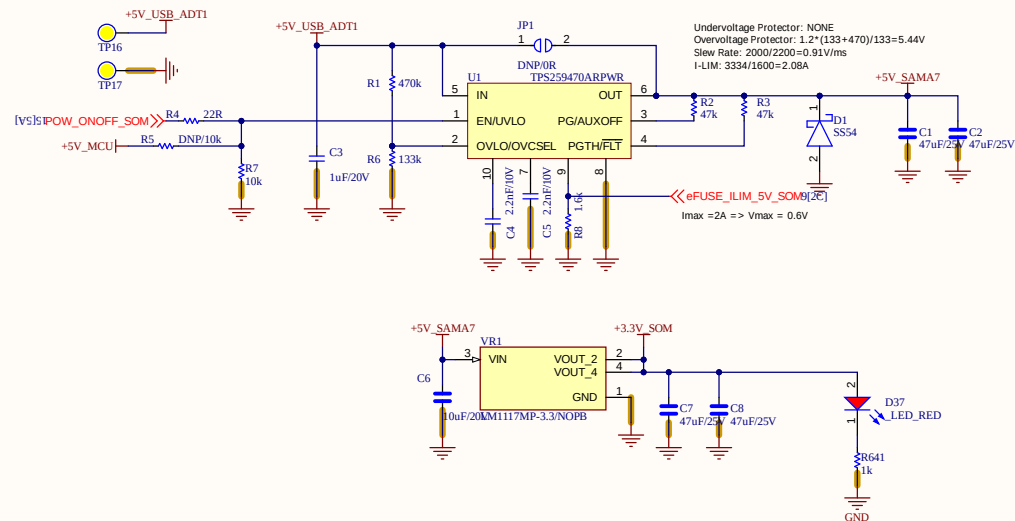
for LASER board



LASER_CHOSE 0: LASER1
1: LASER2
PHOTO_CHOSE 0: PHOTO1
1: PHOTO2



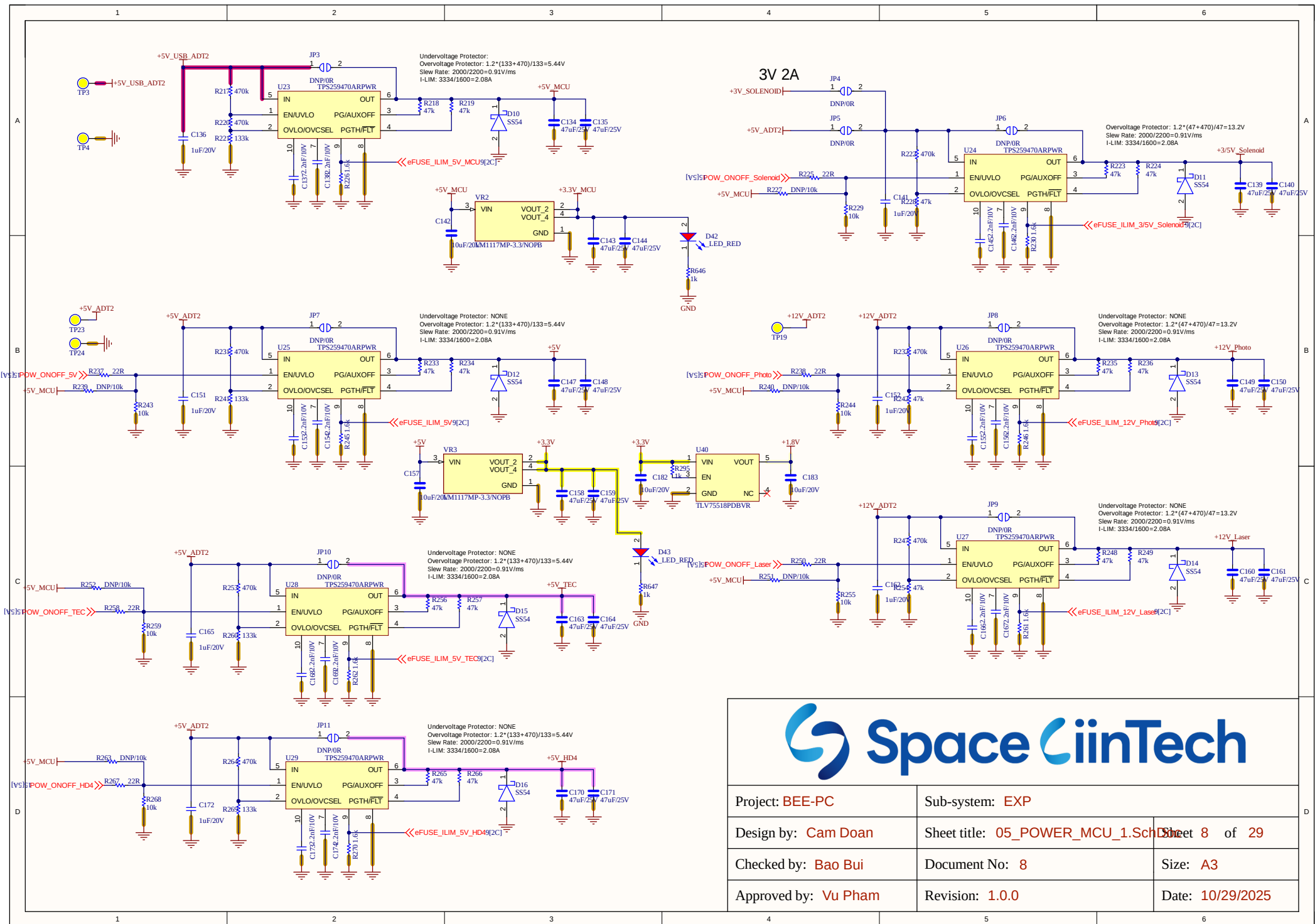
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Design by: Cam Doan	Sheet title: 04_CONNECTOR_OUT_25	Sheet No. of 29
Checked by: Bao Bui	Document No: 6	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

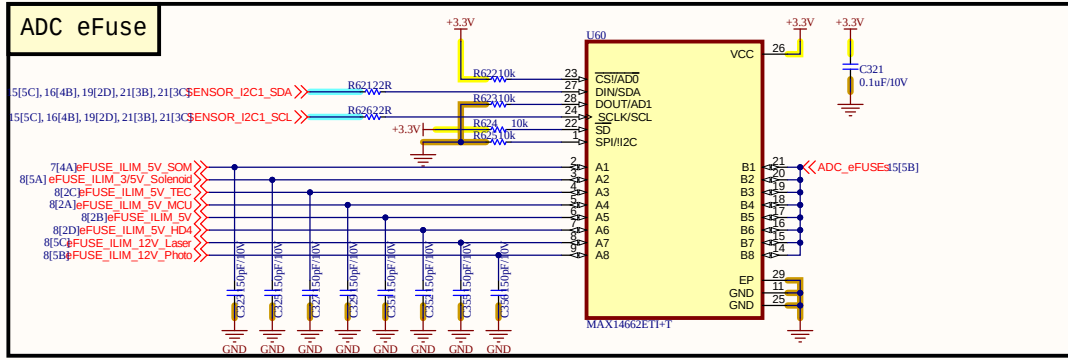
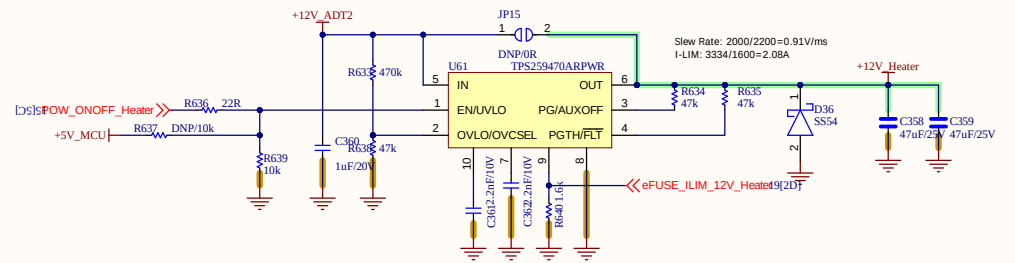
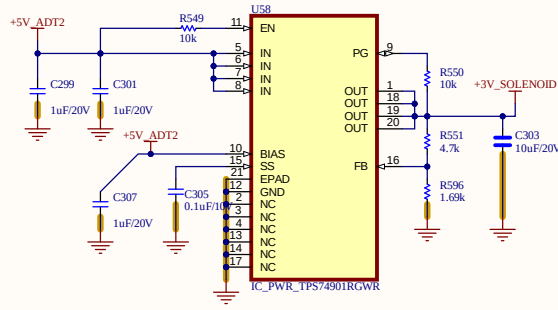


NC	—	2, 3, 4, 13, 14, 17	—	—	No connection. This pin can be left floating or connected to GND to allow better thermal contact to the top-side plane.
----	---	------------------------	---	---	---



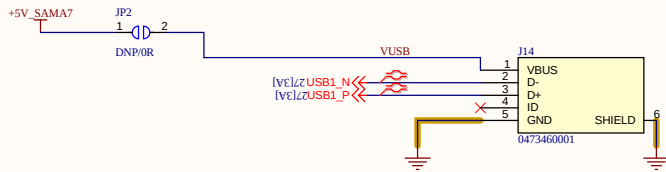
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Design by: Cam Doan	Sheet title: 05_POWER_SAMA7.Doc	Sheet of 29
Checked by: Bao Bui	Document No: 7	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025



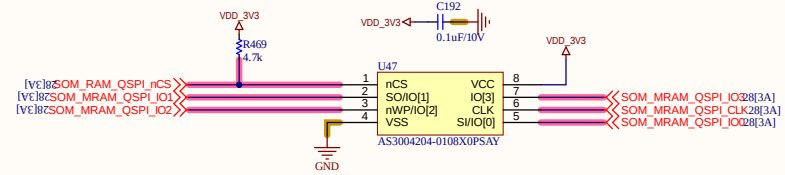


Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 05_POWER_MCU_2.SchDoc	Sheet 9 of 29
Checked by: Bao Bui	Document No: 9	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

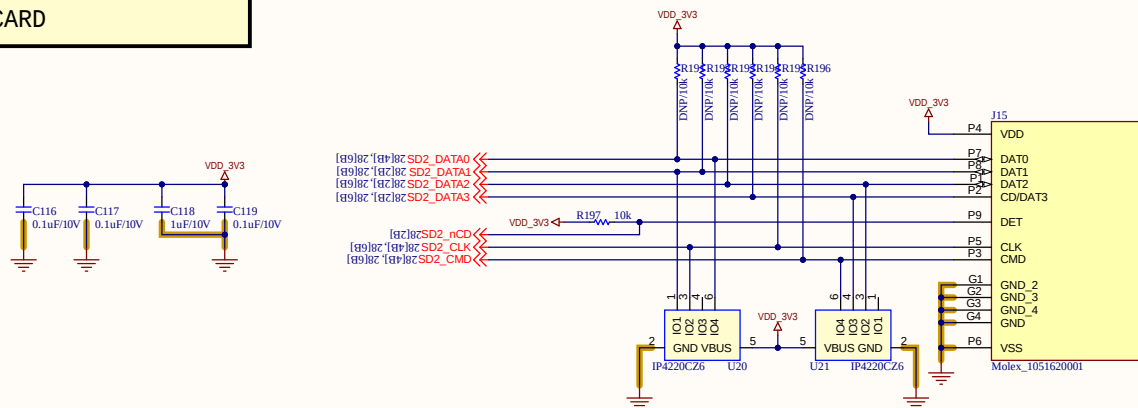
USB



MRAM 4Mbit

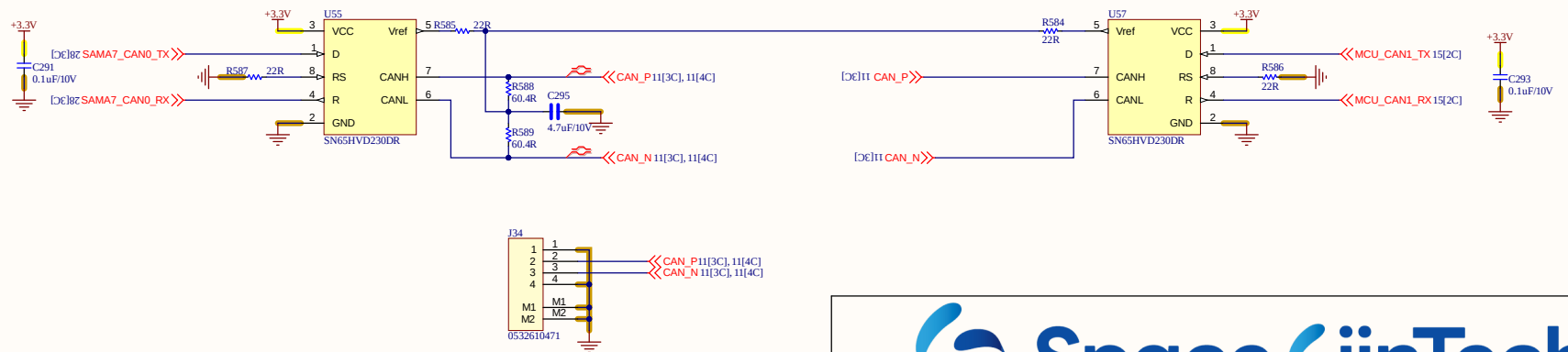
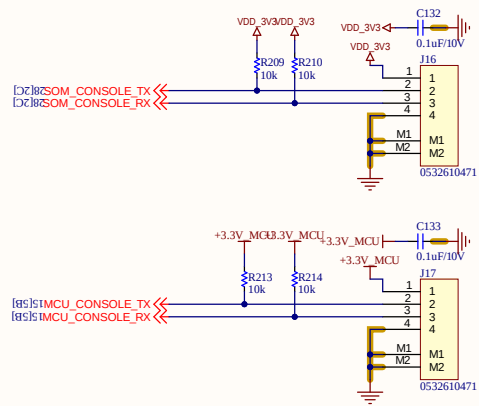


SD CARD



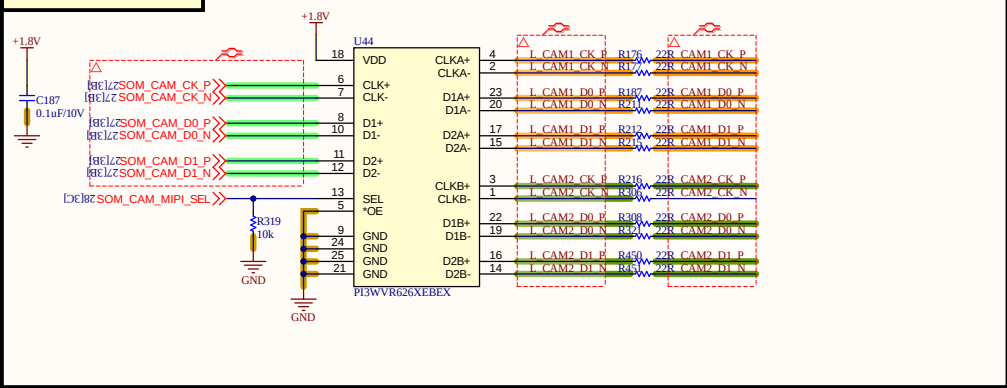
Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 06_SOM_USB_CARD.Sch	Sheet 10 of 29
Checked by: Bao Bui	Document No: 10	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

CONSOLE

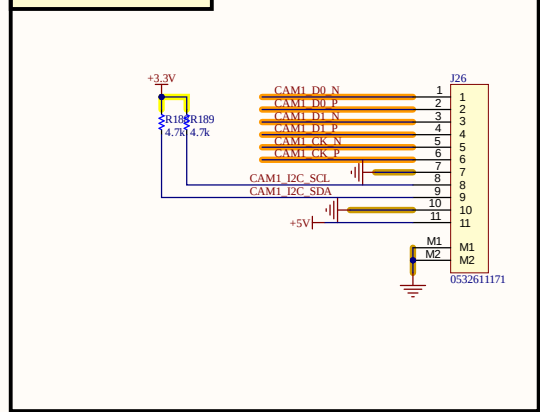


Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 06_SOM_Console.SchDoc	Sheet 11 of 29
Checked by: Bao Bui	Document No: 11	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

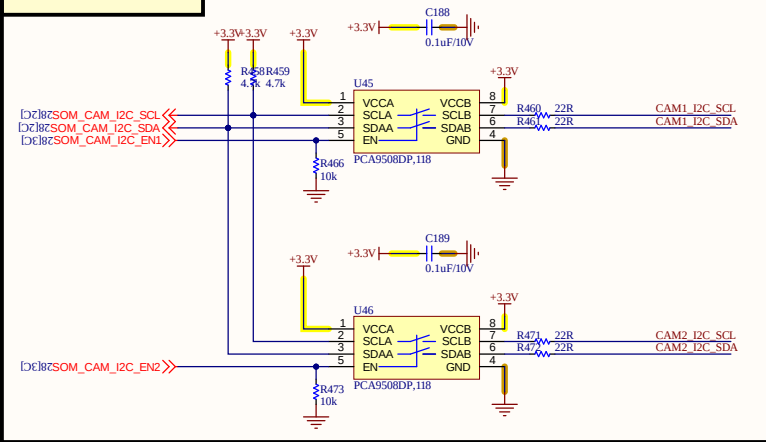
MIPI SWITCH



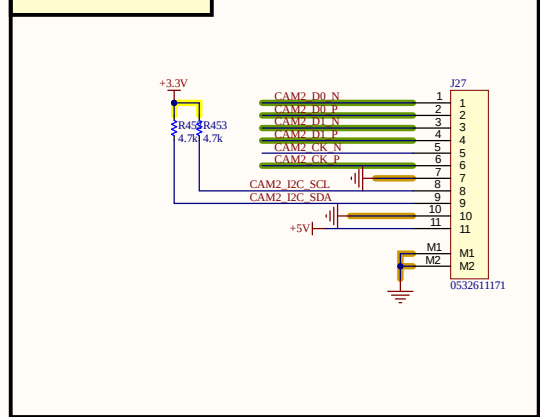
for CAMERA 1



I2C SWITCH

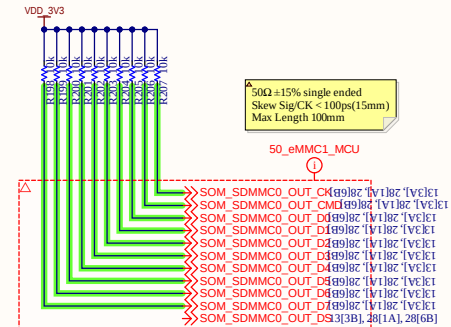
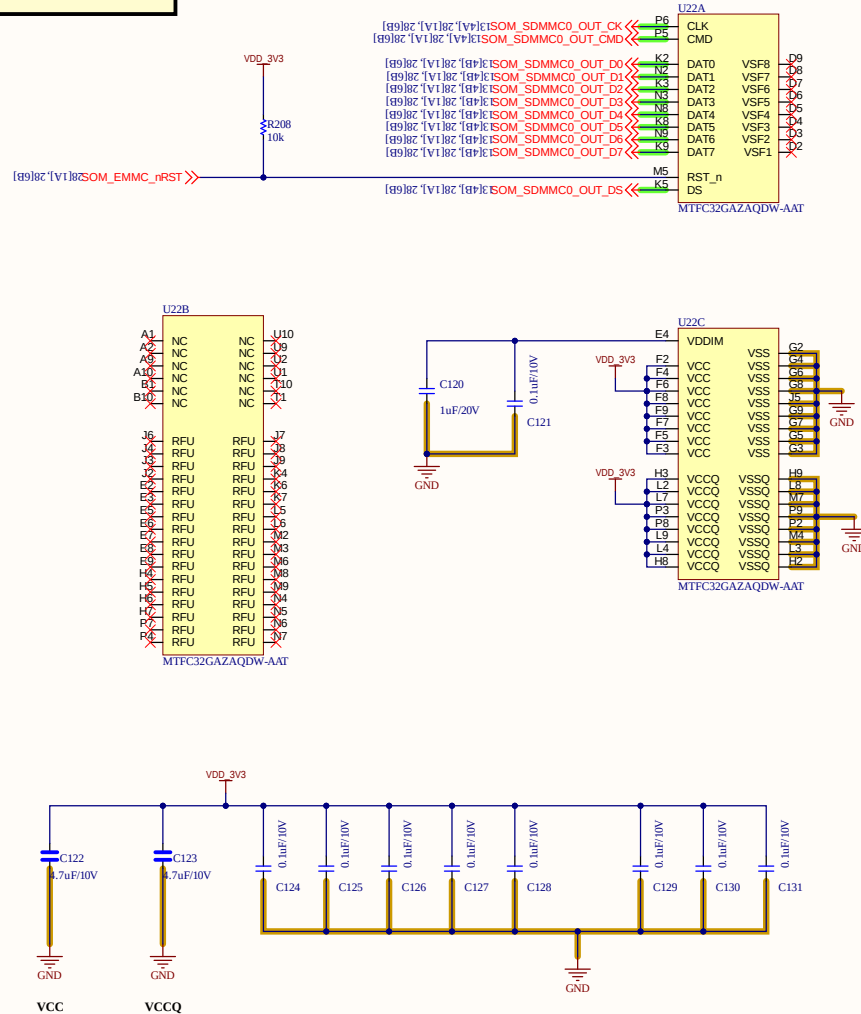


for CAMERA 2



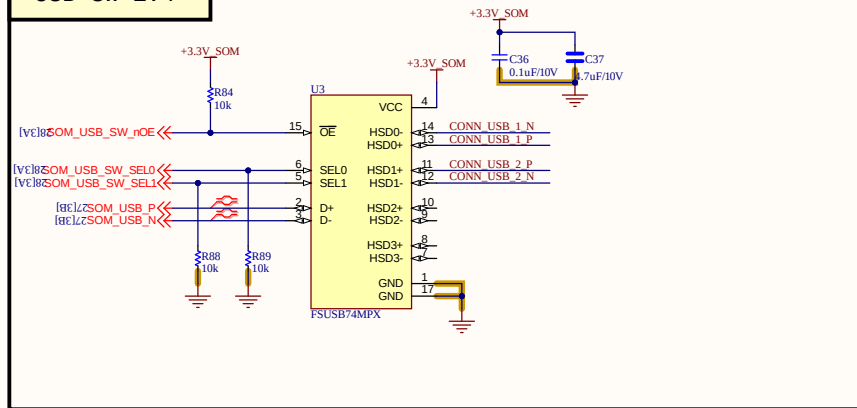
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Design by: Cam Doan	Sheet title: 06_SOM_CAM.SchDoc	Sheet 12 of 29
Checked by: Bao Bui	Document No: 12	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

eMMC - Buffer

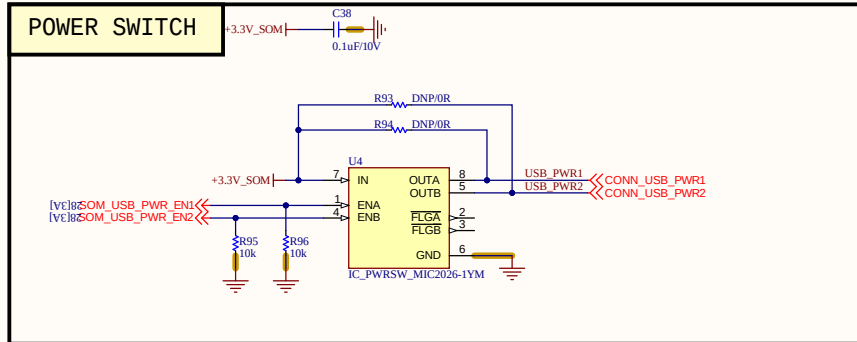


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Design by: Cam Doan	Sheet title: 07_eMMC.SchDoc	Sheet 13 of 29
Checked by: Bao Bui	Document No: 13	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

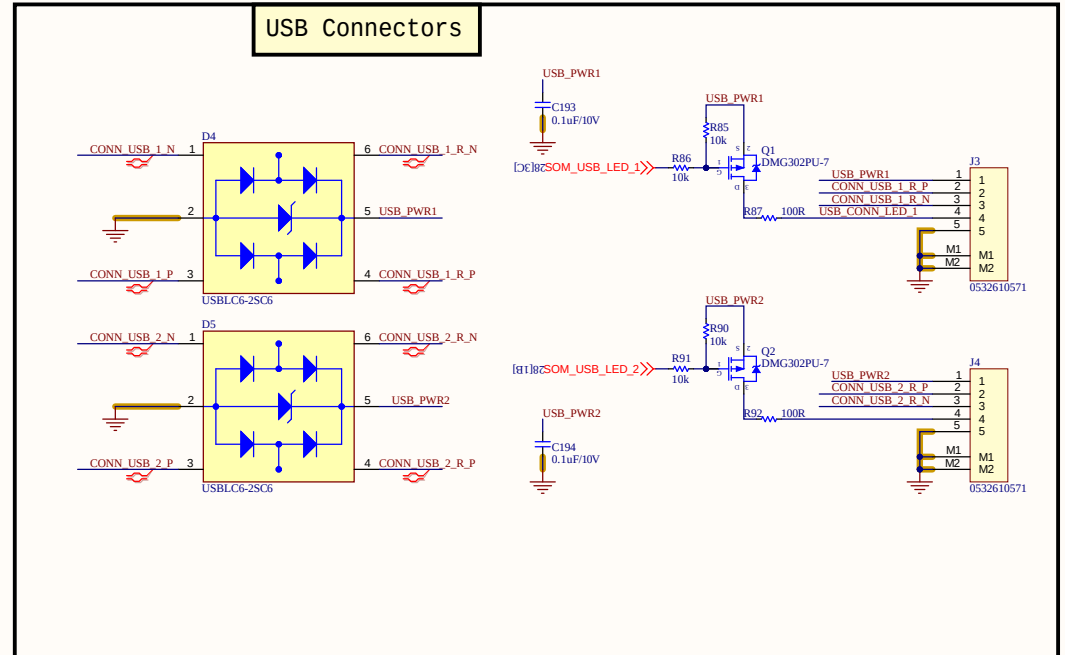
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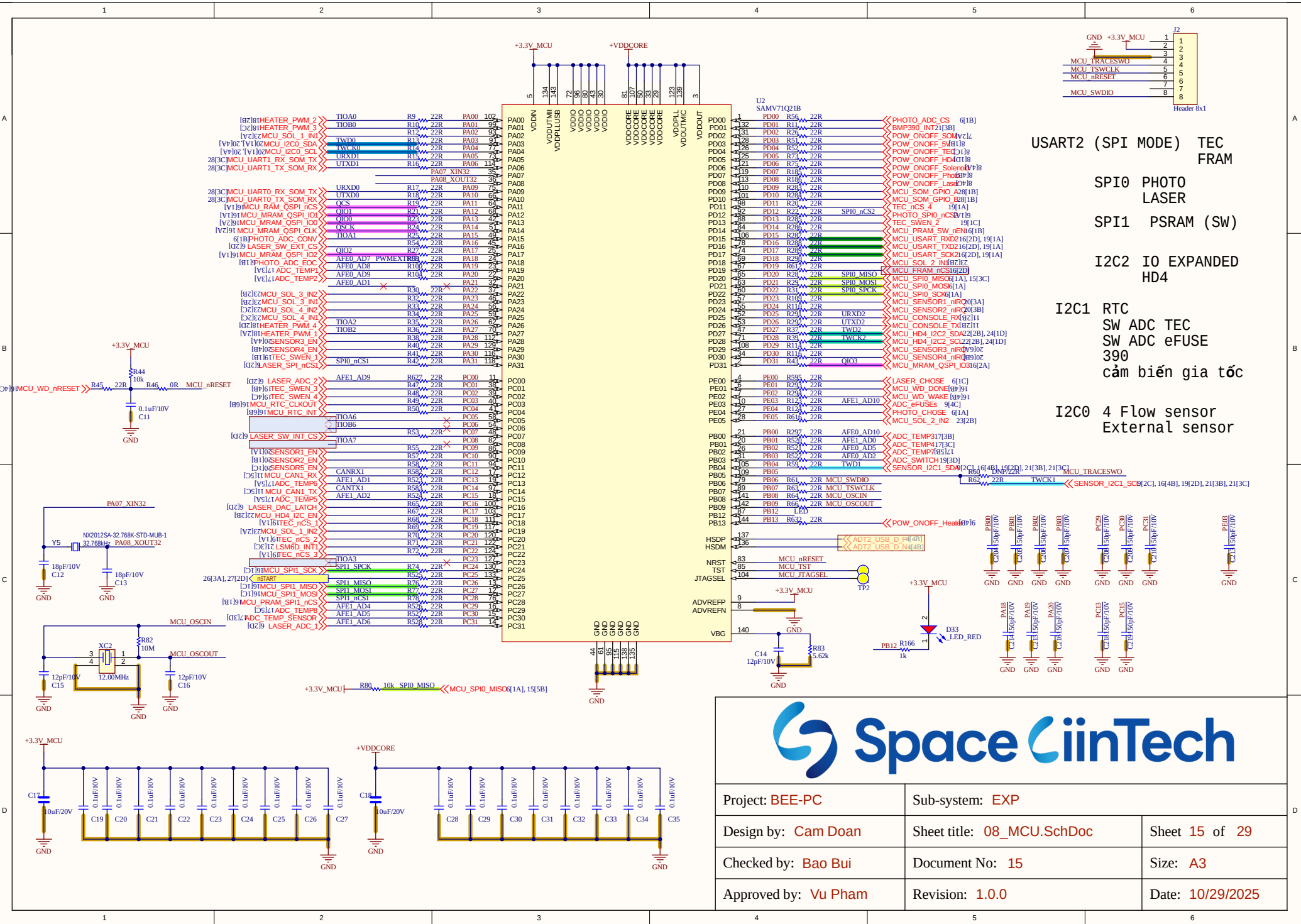
POWER SWITCH



USB Connectors



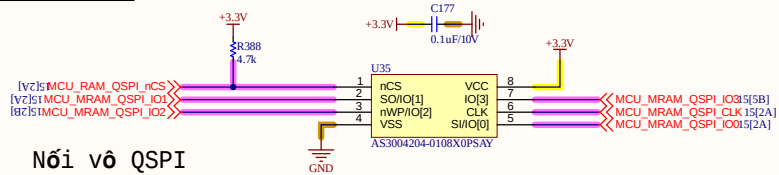
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Design by: Cam Doan	Sheet title: 07_USB.SchDoc	Sheet 14 of 29
Checked by: Bao Bui	Document No: 14	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025



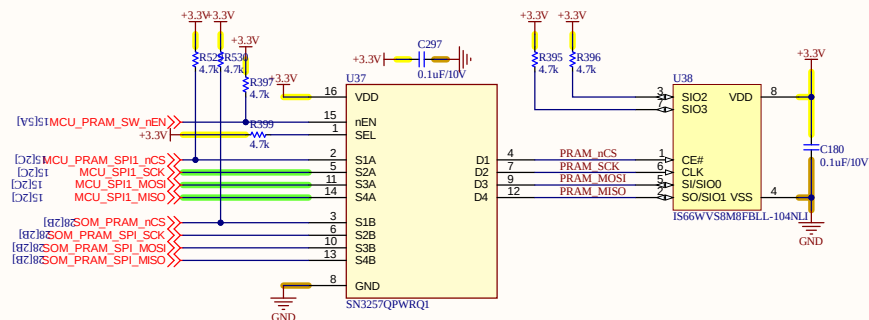
Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 08_MCU.SchDoc	Sheet 15 of 29
Checked by: Bao Bui	Document No: 15	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
V _{DDCORE}	Voltage Range	Full operational	1.7		VDDIN	V

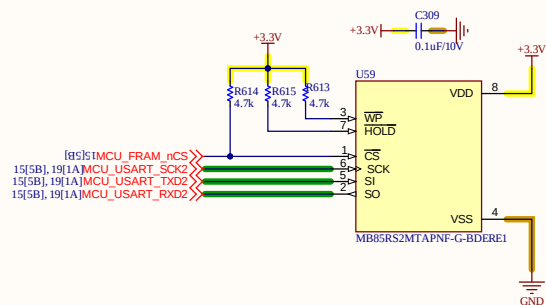
MRAM 4Mbit



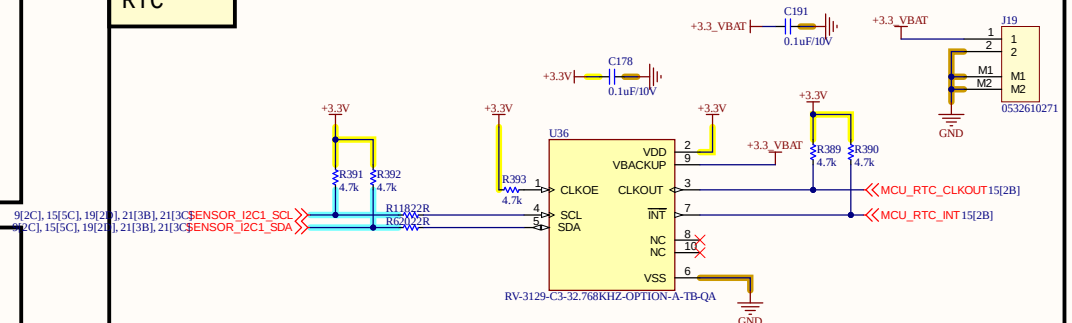
PRAM



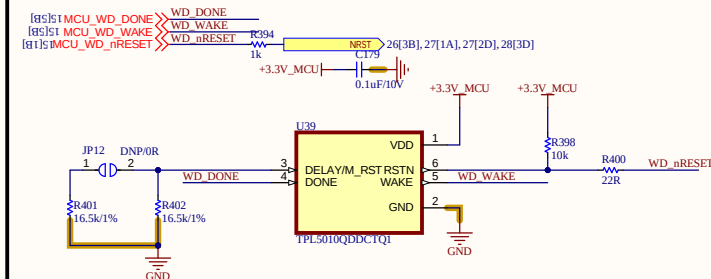
FRAM



RTC

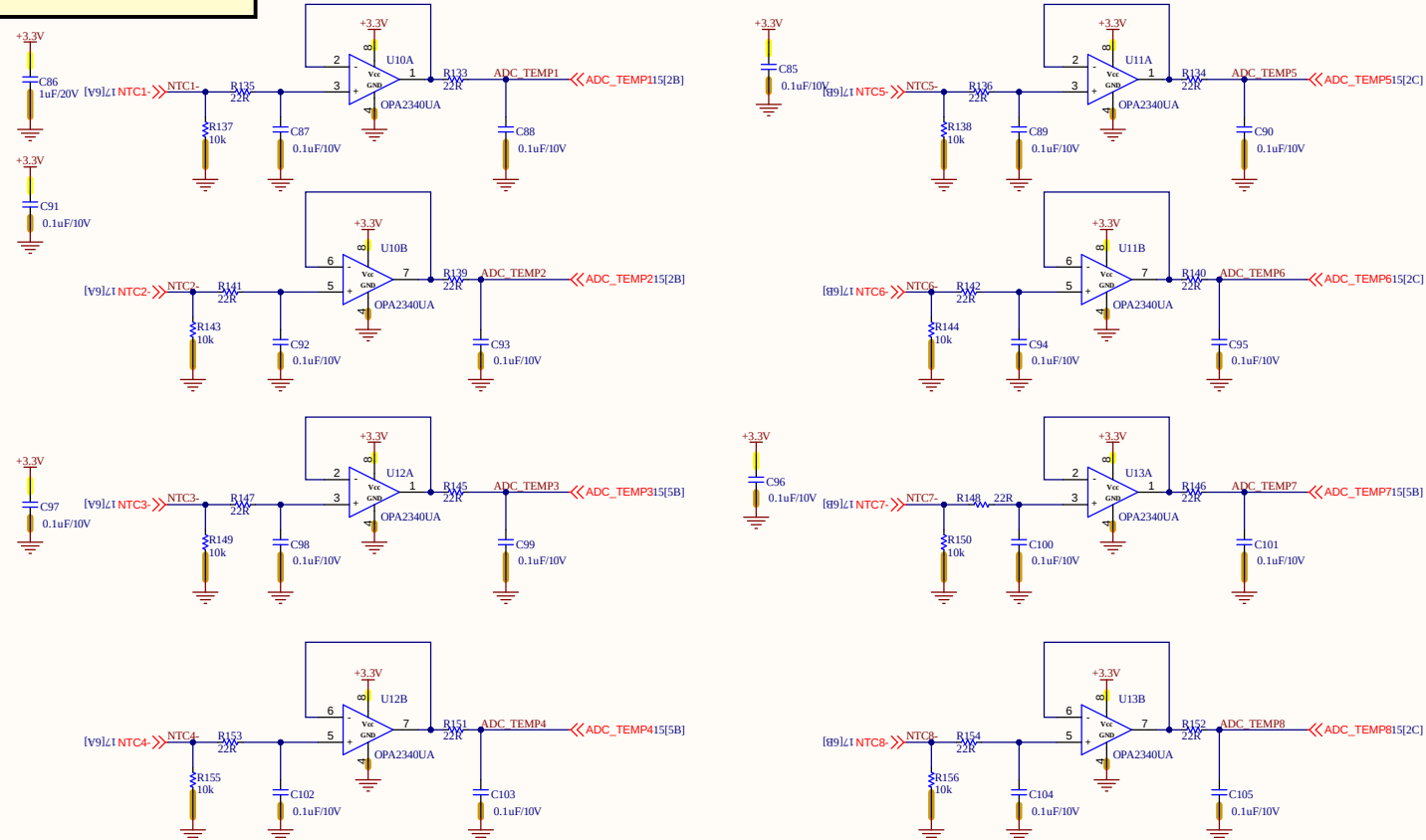


WATCH DOG

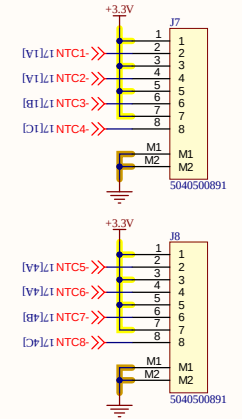


Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 09_MRAM_RTC_WDT_Sch	Sheet 16 of 29
Checked by: Bao Bui	Document No: 16	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

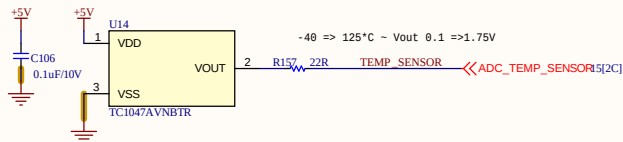
NTC



CONNECTOR FOR NTC

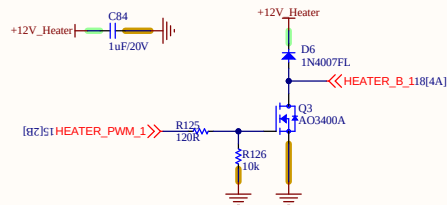


TEMPERATURE SENSOR

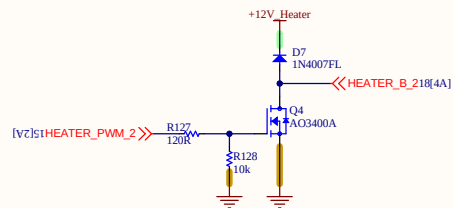


Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 10_NTC_TEMPSENSOR	Sheet 17 of 29
Checked by: Bao Bui	Document No: 17	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

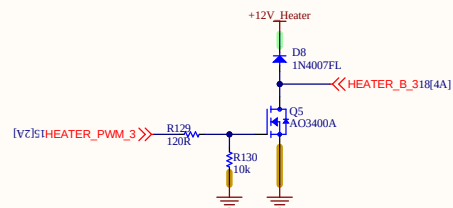
HEATER 1



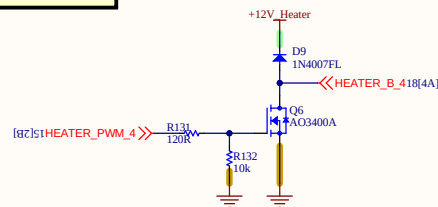
HEATER 2



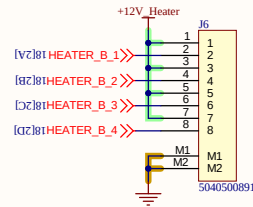
HEATER 3



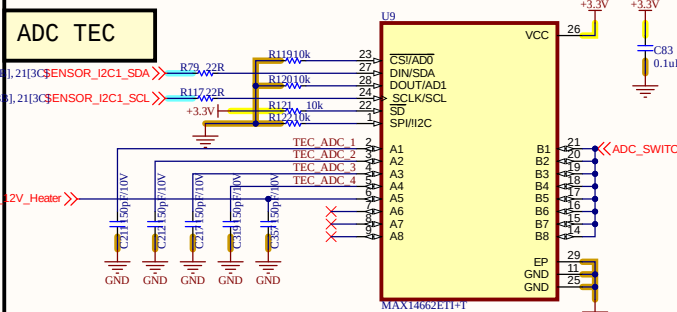
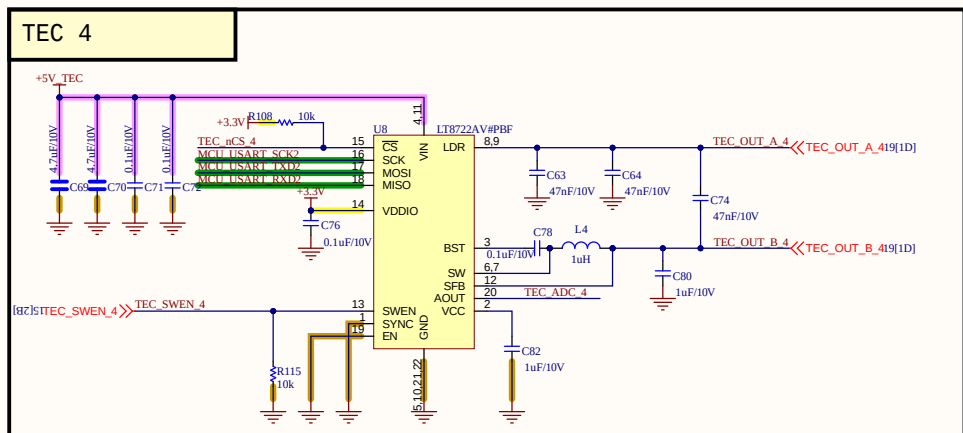
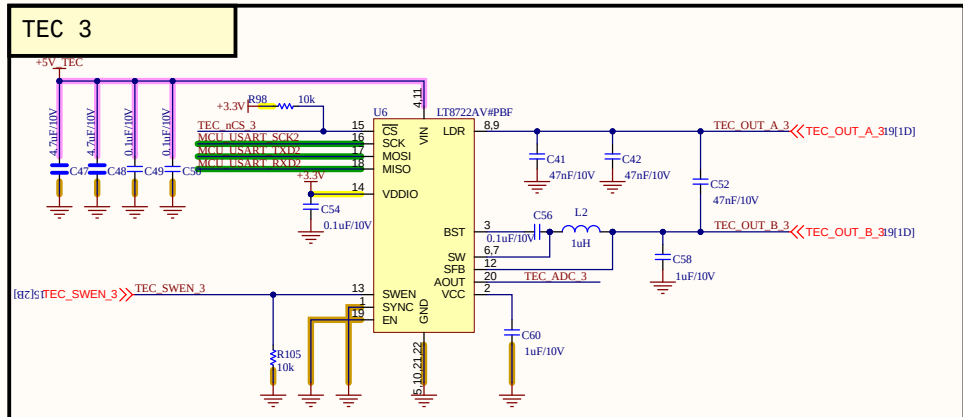
HEATER 4



CONNECTOR FOR NTC

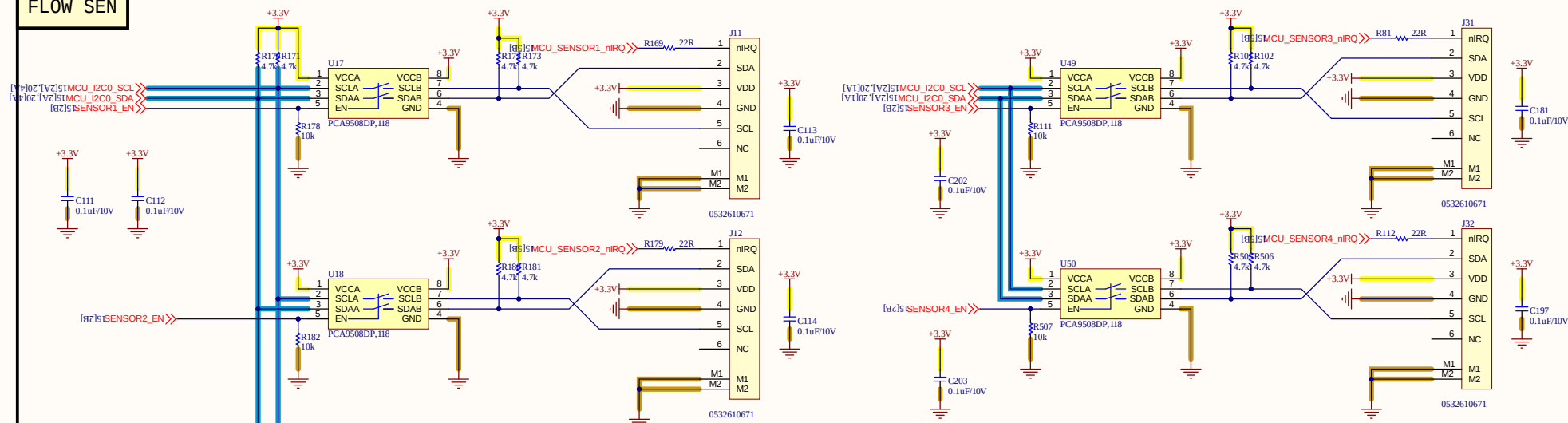


Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 11_HEATER.SchDoc	Sheet 18 of 29
Checked by: Bao Bui	Document No: 18	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

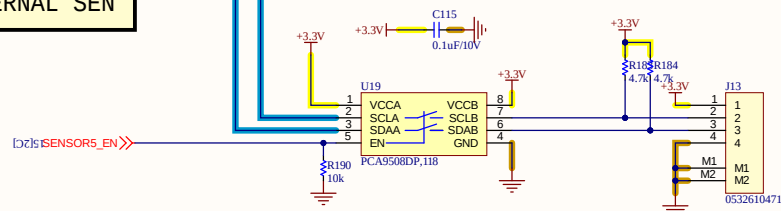


Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 12_TEC.SchDoc	Sheet 19 of 29
Checked by: Bao Bui	Document No: 19	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

FLOW SEN

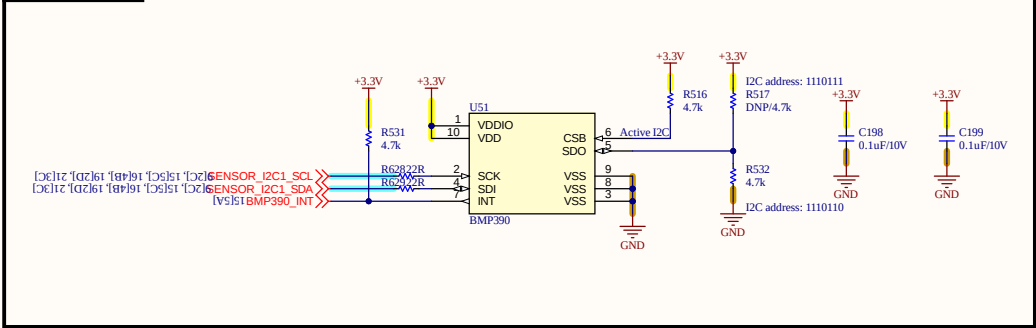


EXTERNAL SEN

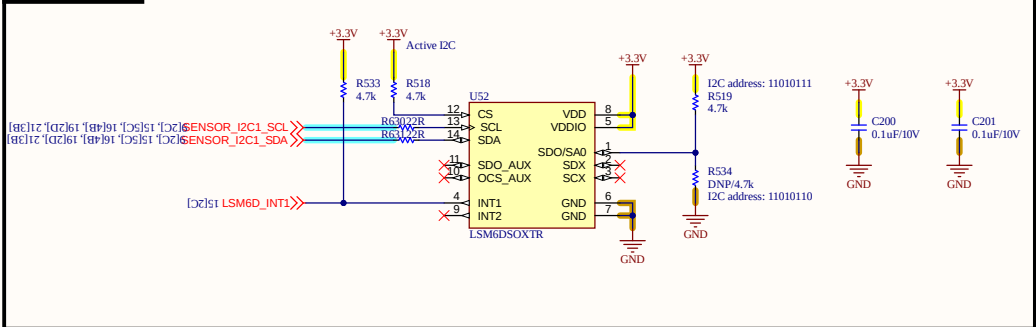


Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 13_FLOW_Sensor.SchDoc	
Checked by: Bao Bui	Document No: 20	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

BMP390



LSM6DSOX



Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 13_Sensors_Onboard.Sch	Sheet 21 of 29
Checked by: Bao Bui	Document No: 21	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

6.3. Connection diagram PC

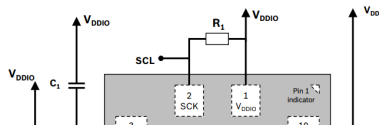
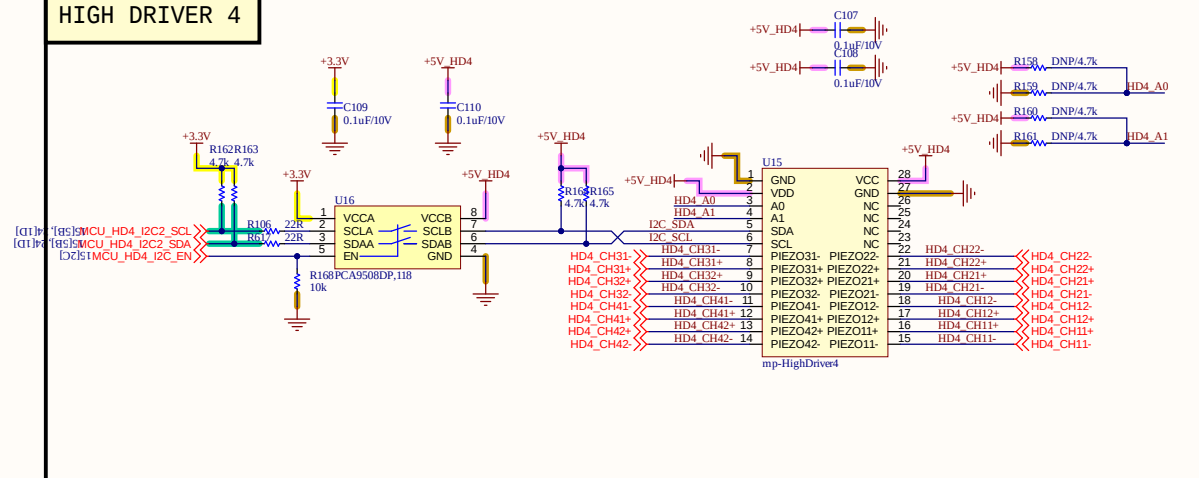
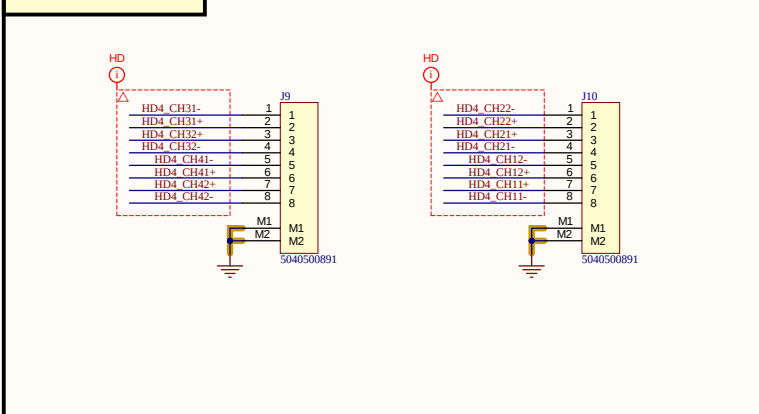


Figure 24. LSM6DSOX electrical connections in mode 1

HIGH DRIVER 4

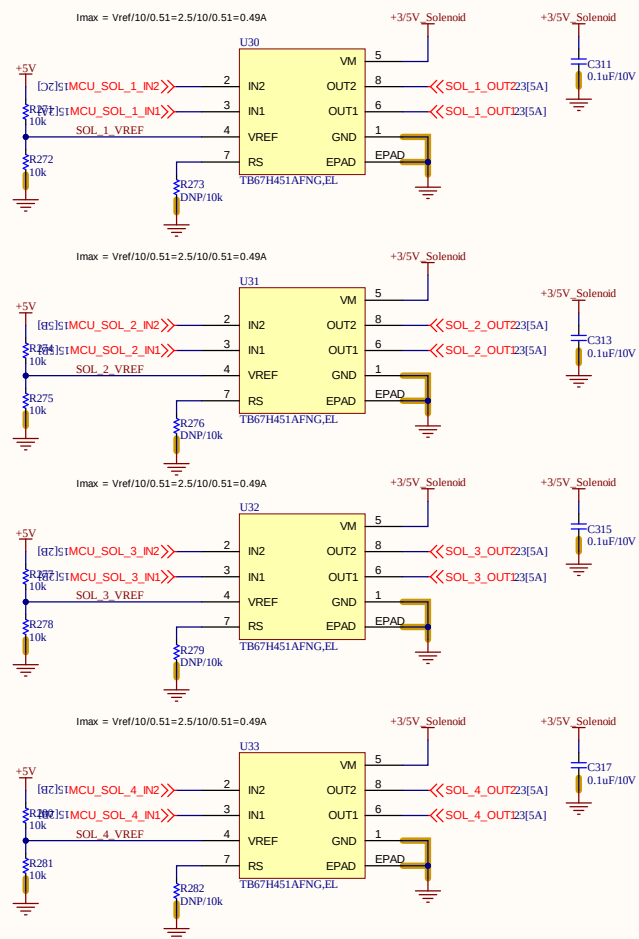


CONNECTOR BUMP

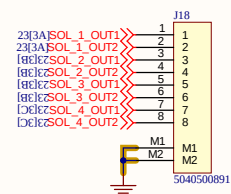


Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 14_HD4_PUMP.SchDoc	Sheet 22 of 29
Checked by: Bao Bui	Document No: 22	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

SOLENOID



CONNECTOR SOLENOID



Project: BEE-PC

Sub-system: EXP

Design by: Cam Doan

Sheet title: 15_SOLEDNOID.SchDoc Sheet 23 of 29

Checked by: Bao Bui

Document No: 23

Size: A3

Approved by: Vu Pham

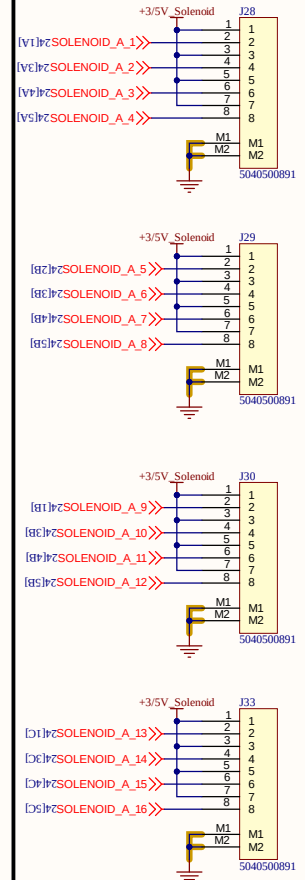
Revision: 1.0.0

Date: 10/29/2025

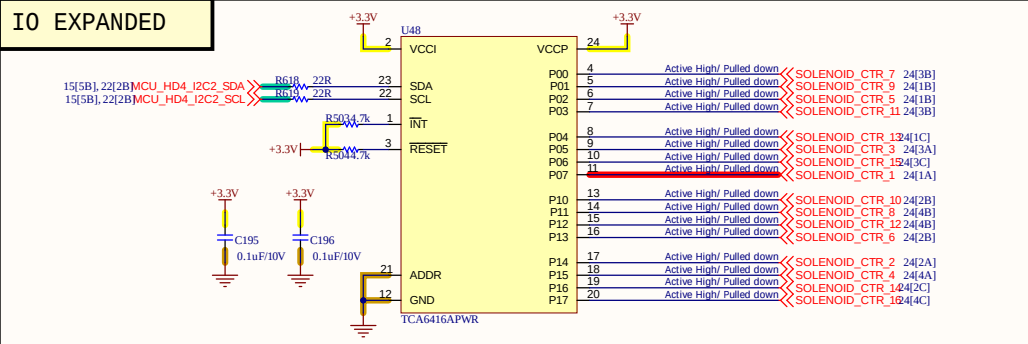
SOLENOID x16



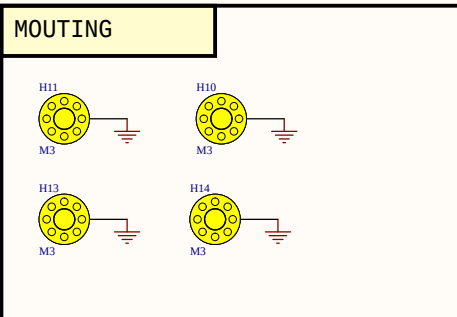
CONN SOLENOID



IO EXPANDED

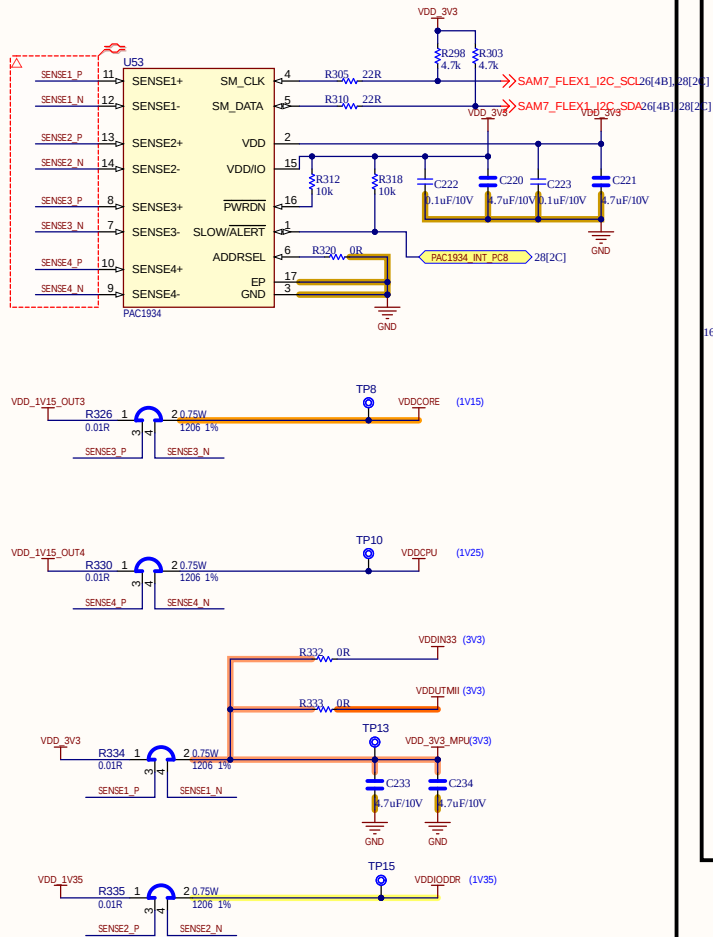


Project: BEE-PC	Sub-system: EXP
Design by: Cam Doan	Sheet title: 15_SOLENOID 2.SchDoc
Checked by: Bao Bui	Document No: 24
Approved by: Vu Pham	Revision: 1.0.0
	Size: A3
	Date: 10/29/2025

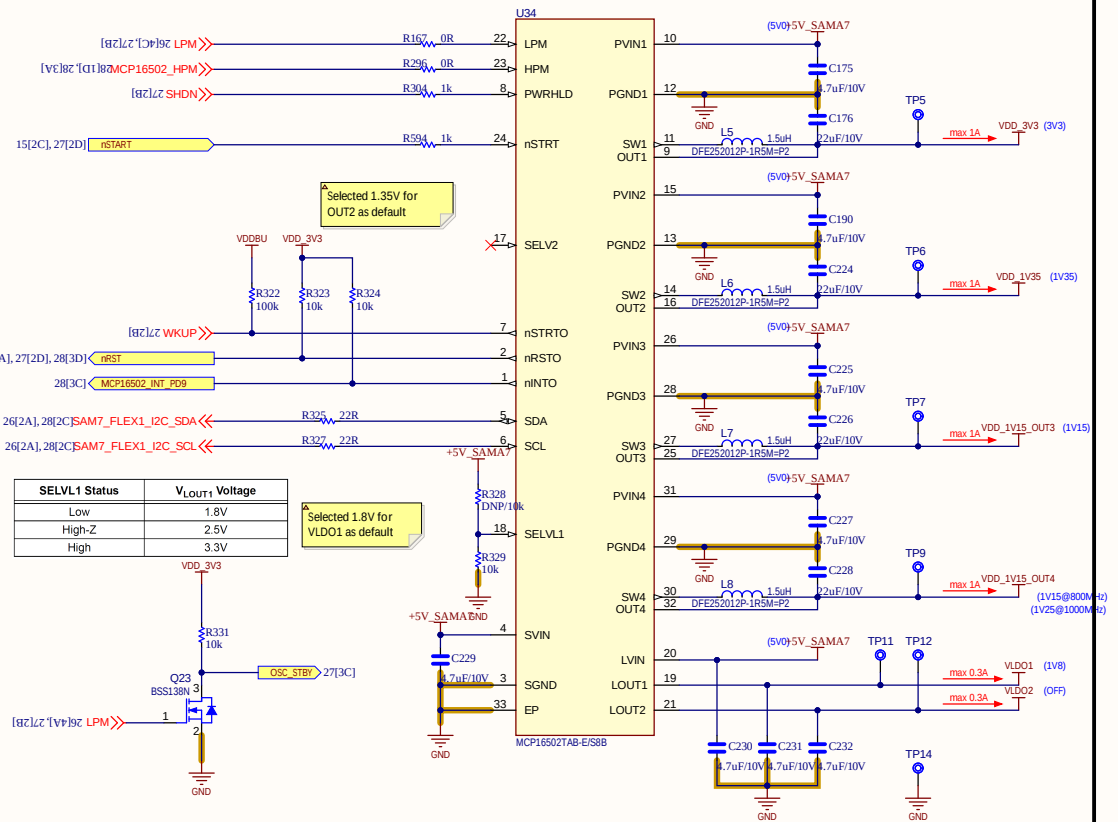


Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: MOUTING.SchDoc	Sheet 25 of 29
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Power Measurement and Distribution



PMIC

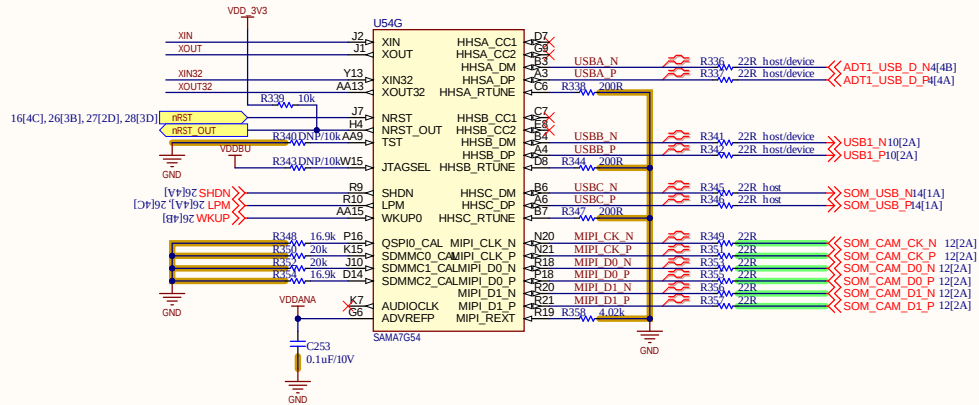


Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 06_SAMA7G54-PMIC.Sch	Sheet 26 of 29
Checked by: Bao Bui	Document No: 26	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

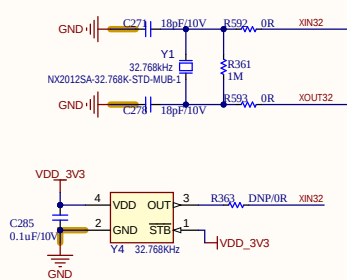
Thêm CAN giữa 2 khối

MPU configuration

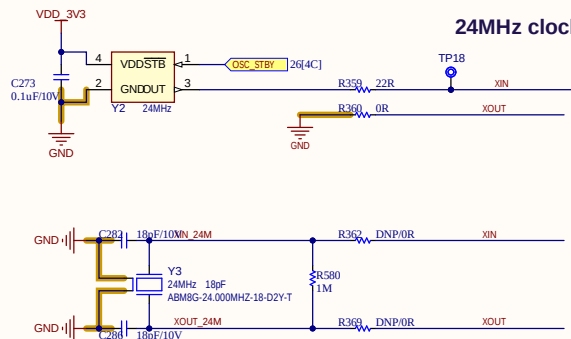
MPU decoupling



32.768KHz clock



24MHz clock



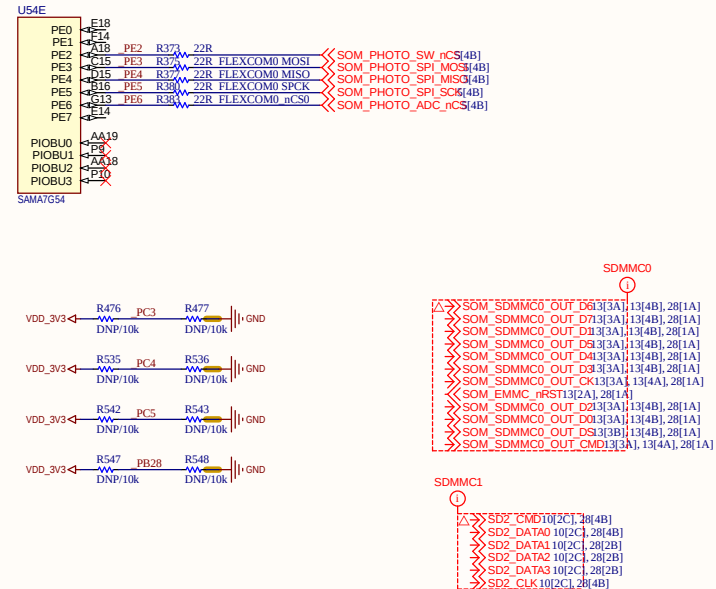
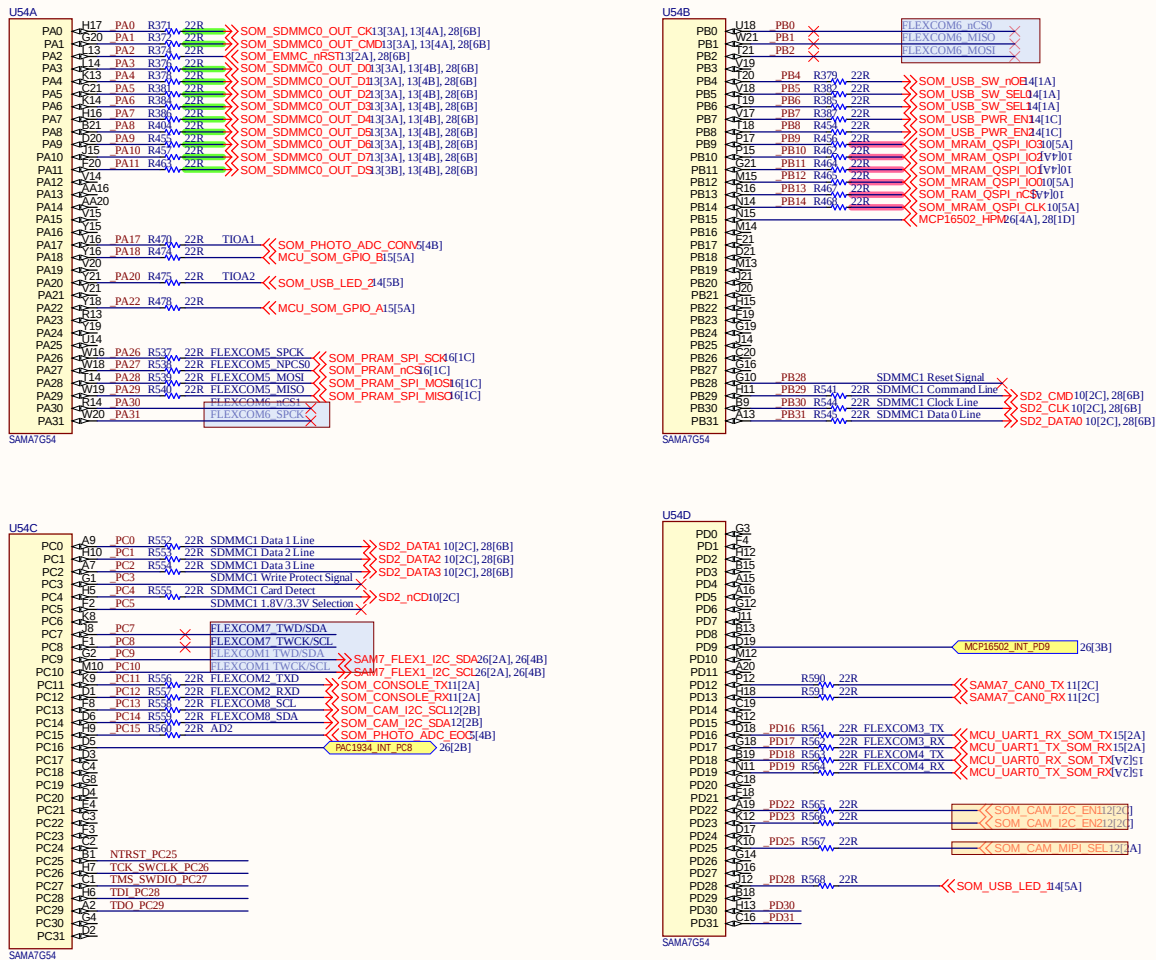
+3.3_VBAT

VDDBAT



Project: BEE-PC	Sub-system: EXP	
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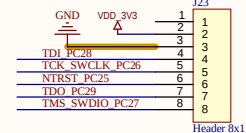
SAMA7G5 Bus Distribution



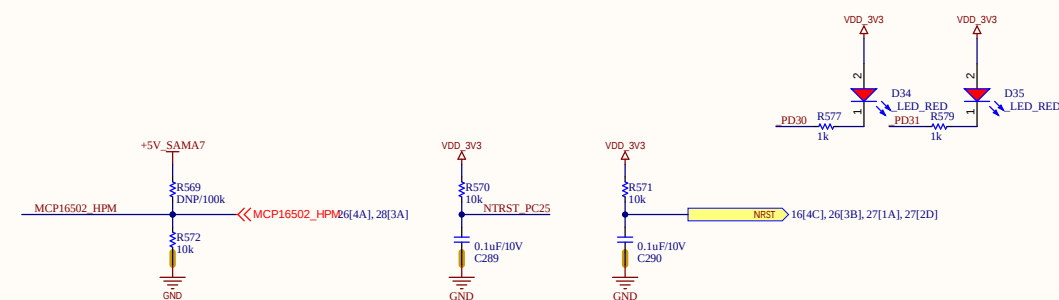
SPI:3 F0: PE3456 PHOTO
 F5: PA26-789 PRAM (SW)
 F6

UART:3 F4:PD18 PD19 SOM
 F3:PD16 PD17 SOM
 F2:PC11 12 CONSOLE

I2C:2 F1:PC9 10 PMIC
 F8:PC1314 CAMERA
 F7:PC7 8 NG



Project: BEE-PC	Sub-system: EXP	
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BEE-PC 4U EXP V1.0.0

Designed by: Cam Doan
Checked by: Bao Bui
Approved by: Vu Pham

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Board Stack Report